

Key features of parabolas



- 1
- | | | | | | | | |
|---|---------|---|---------|---|----------|---|----------|
| a | 5 units | b | 5 units | c | 10 units | d | 10 units |
|---|---------|---|---------|---|----------|---|----------|

- 2
- | | | | | | | | |
|---|---------|---|---------|---|----------|---|----------|
| a | 5 units | b | 5 units | c | 10 units | d | 10 units |
|---|---------|---|---------|---|----------|---|----------|

- 3
- | | | | | |
|---|-----------------------------------|---------------|-----------------------|------------------------------------|
| a | i Upwards
v $y = 1$ | ii $(2, 4)$ | iii $a = 3$ | iv $(2, 7)$ |
| b | i Upwards
v $y = -\frac{7}{2}$ | ii $(-4, -2)$ | iii $a = \frac{3}{2}$ | iv $\left(-4, -\frac{1}{2}\right)$ |
| c | i Downwards
v $y = 2$ | ii $(3, -2)$ | iii $a = 4$ | iv $(3, -6)$ |
| d | i Upwards
v $y = \frac{11}{4}$ | ii $(0, 3)$ | iii $a = \frac{1}{4}$ | iv $\left(0, \frac{13}{4}\right)$ |
| e | i Upwards
v $y = -4$ | ii $(0, 0)$ | iii $a = 4$ | iv $(0, 4)$ |
| f | i Upwards
v $y = -4$ | ii $(-2, 0)$ | iii $a = 4$ | iv $(-2, 4)$ |

4



- a**
- i** To the right **ii** $(0, 3)$ **iii** $a = 3$ **iv** $(3, 3)$
 - v** $x = -3$
- b**
- i** To the right **ii** $(2, 0)$ **iii** $a = 4$ **iv** $(6, 0)$
 - v** $x = -2$
- c**
- i** To the right **ii** $(2, 3)$ **iii** $a = 4$ **iv** $(6, 3)$
 - v** $x = -2$
- d**
- i** To the left **ii** $(2, -4)$ **iii** $a = 3$ **iv** $(-1, -4)$
 - v** $x = 5$

5

- a**
- i** $(0, 0)$ **ii** $x = 0$ **iii** $a = 4$ **iv** $(0, 4)$
 - v** $y = -4$
- b**
- i** $(0, 0)$ **ii** $x = 0$ **iii** $a = \frac{1}{12}$ **iv** $\left(0, -\frac{1}{12}\right)$
 - v** $y = \frac{1}{12}$
- c**
- i** $(0, 0)$ **ii** $y = 0$ **iii** $a = 5$ **iv** $(-5, 0)$
 - v** $x = 5$
- d**
- i** $(0, 0)$ **ii** $y = 0$ **iii** $a = \frac{1}{20}$ **iv** $\left(-\frac{1}{20}, 0\right)$
 - v** $x = \frac{1}{20}$

6

- a** $(2, -1)$ **b** $(2, 1)$ **c** $y = -3$ **d** $x = 2$
- e** $(-\infty, \infty)$ **f** $[-1, \infty)$

7

a $(-1, 3)$

b $(1, 3)$

e $[-1, \infty)$

f $(-\infty, \infty)$



c $x = -3$

d $y = 3$

8

a $x = \pm 2\sqrt{6}$

b $(2\sqrt{6}, -7)$ and $(-2\sqrt{6}, -7)$

9

a $y = \pm 2\sqrt{15}$

b $(-7, 2\sqrt{15})$ and $(-7, -2\sqrt{15})$

10 $k = -2, -6$

Graphs of parabolas

11

a Downwards

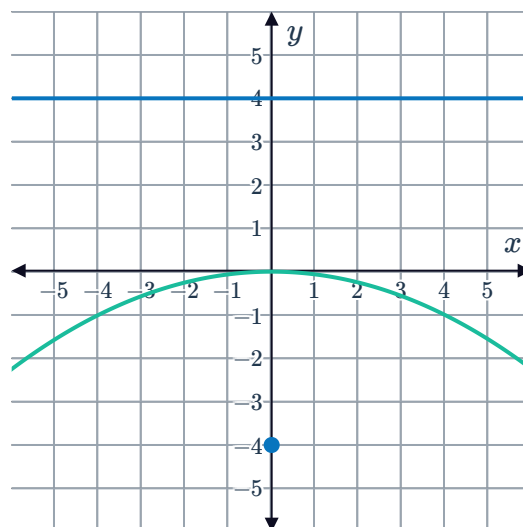
c $a = 4$

e $y = 4$

b $(0, 0)$

d $(0, -4)$

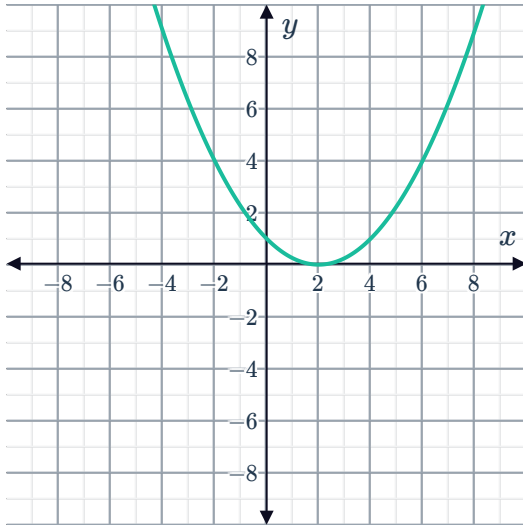
f



12

a $(0,1)$

c



e $(2,1)$



b $y = -1$

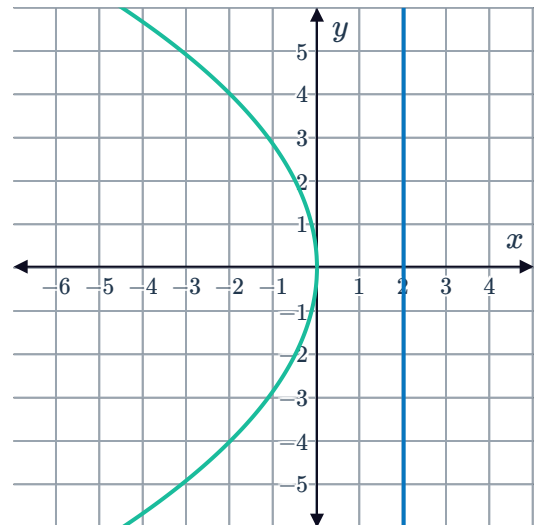
d $(x-2)^2 = 4y$

f $y = -1$

13

a $x = 2$

b



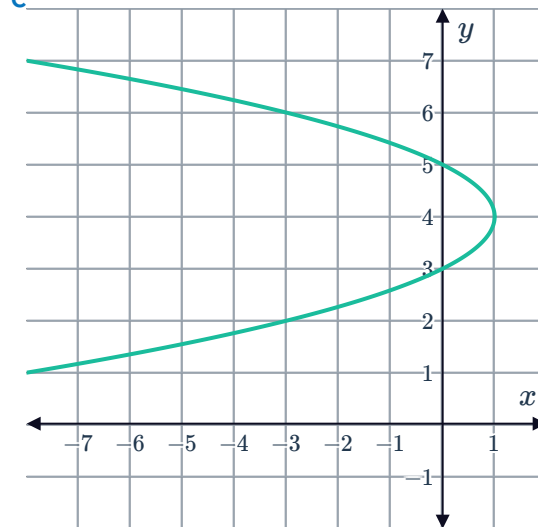
c $y = \pm 4$

d $(-2,4)$ and $(-2,-4)$

14

a To the left.

b $(1, 4)$



15

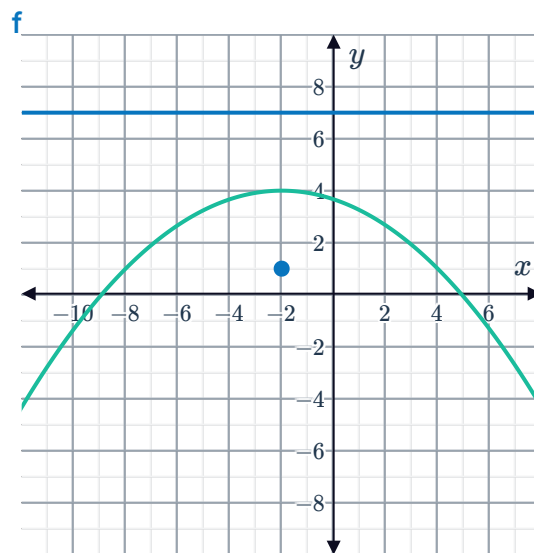
a Downwards

b $(-2, 4)$

c $a = 3$

d $(-2, 1)$

e $y = 7$



16

a Upwards

c $a = \frac{3}{2}$

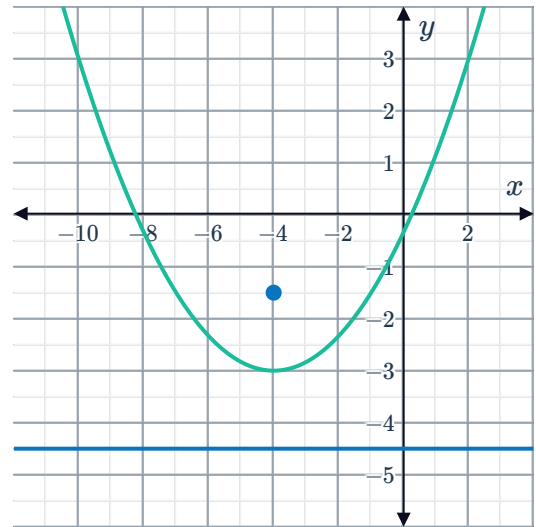
e $y = -\frac{9}{2}$



b $(-4, -3)$

d $\left(-4, -\frac{3}{2}\right)$

f



g $\frac{15}{2}$ units

Equations of parabolas

17

a $x^2 = 16y$

b $x^2 = 12y$

c $y^2 = -24x$

d $y^2 = -2x$

e $x^2 = 2y$

f $x^2 = \frac{49}{5}y$

g $(x - 6)^2 = 16y$

h $(y - 8)^2 = 20x$

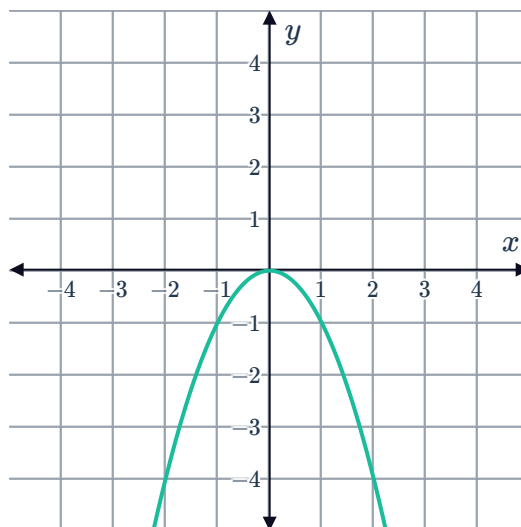
18 $y^2 = \pm 16x$

19

a $x^2 = -y$

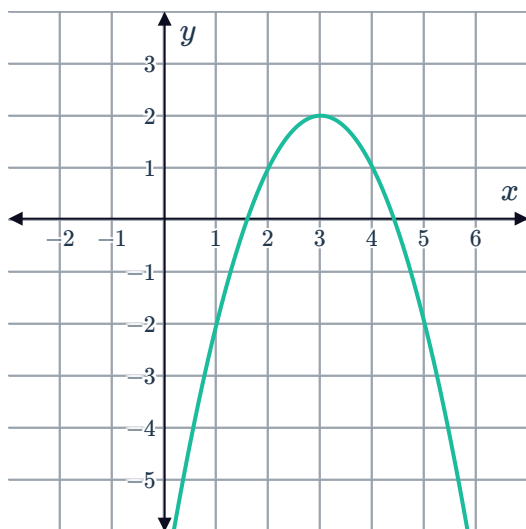


b



20

a



b $(x - 3)^2 = -(y - 2)$

Applications

21

a $s = 12$

b 337

22

a $t = 3$

b 122